

A Finite Element Analysis of Second-Order Elliptic Boundary Value Problems

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Motivation

Physical Problems & Partial Differential Equations

Modeling of several physical phenomena are governed by partial differential equations.

Limitations of Analytical Methods

Finding exact solutions of the governing PDEs by analytical methods is typically difficult due to many complexities.

The Need for Numerical Solutions

Infeasibility of finding the exact solutions creates a need to study numerical solutions of PDEs.

Goal

Weak Formulation & Galerkin Approximation

To study the weak solution of PDEs and Galerkin Approximations in the Sobolev space framework.

Finite Element Methods

To discuss the finite element method for solving second order problems rigorously, along with error analysis.

Domain Geometry & FEM

To explore the limitations of FEM on domain geometry via rate of convergence of the approximate solutions.

Sobolev Space

Let $m > 0$ be an integer and let $1 \leq p \leq \infty$. The Sobolev Space $W^{m,p}(\Omega)$ is defined by $W^{m,p}(\Omega) = \{u \in L^p(\Omega) | D^\alpha u \in L^p(\Omega) \text{ for all } |\alpha| \leq m\}$.

For $p = 2$, we denote the space as $H^m(\Omega)$, which is a Hilbert space.

The closure of the subspace $\mathcal{D}(\Omega)$ in $W^{m,p}(\Omega)$ is the closed subspace $W_0^{m,p}(\Omega)$. When $p = 2$, we denote this subspace by $H_0^m(\Omega)$.

Poincaré Inequality

Let Ω be a bounded open set in $\mathbb{R}^N$ and let $1 \leq p < \infty$. Then, there exists a constant $C = C(\Omega, p) > 0$ such that,

$$\|u\|_{L^p(\Omega)} \leq C \|\nabla u\|_{L^p(\Omega)} \quad \forall u \in W_0^{1,p}(\Omega).$$

Preliminaries (Continued)

Continuous & H -elliptic Bilinear Form

Let H be a real Hilbert space. A bilinear form $a : H \times H \rightarrow \mathbb{R}$ is said to be

- **continuous** if $\exists M > 0$ such that $|a(u, v)| \leq M \|u\| \|v\| \quad \forall u, v \in H$.
- **H -elliptic** if $\exists \alpha > 0$ such that $a(v, v) \geq \alpha \|v\|^2 \quad \forall v \in H$.

Example

For $H = \mathbb{R}^N$, $u = (u_1, u_2, \dots, u_N)$, $v = (v_1, v_2, \dots, v_N)$ in $\mathbb{R}^N$, define

$$a(u, v) := v^T A u, \text{ where } A = (a_{ij})_{N \times N}; i, j \in \{1, 2, \dots, N\}.$$

- By Cauchy-Schwartz inequality, a is continuous.
- If A is positive definite and symmetric matrix, we have the H -ellipticity.

The Homogeneous Dirichlet Problem

Lax-Milgram Theorem

Let H be a Hilbert space and $V \subseteq H$ be a closed subspace. Let $a(\cdot, \cdot)$ be a continuous, H -elliptic bilinear form on H . Then for any $f \in H$, there exists a unique $u \in V$ such that,

$$a(u, v) = \langle f, v \rangle \quad \forall v \in V.$$

Homogeneous Boundary Condition

Consider the homogeneous Dirichlet problem,

$$\begin{aligned} -\Delta u &= f \quad \text{in } \Omega, \\ u &= 0 \quad \text{on } \partial\Omega, \end{aligned}$$

where $f \in L^2(\Omega)$ be a given function.

The Homogeneous Dirichlet Problem (Continued)

Weak Formulation

Find $u \in H_0^1(\Omega)$ such that,

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega).$$

- Existence and uniqueness is ensured using Lax-Milgram theorem.
- Now, we prove the existence and uniqueness using Galerkin method.

Galerkin Approximation

Objective

To prove the existence and uniqueness of the solution to the problem: Find $u \in H_0^1(\Omega)$ such that $a(u, v) = \langle f, v \rangle \quad \forall v \in H_0^1(\Omega)$,

where $\langle f, v \rangle = \int_{\Omega} f v \, dx$, $f \in L^2(\Omega)$, and,

$$a(u, v) = \int_{\Omega} \nabla u \cdot \nabla v \, dx \quad \forall v \in H_0^1(\Omega).$$

Galerkin Approximation (Continued)

Sketch of the Procedure

- To find a suitable finite dimensional subspace, $V_m = \text{span}\{e_1, e_2, \dots, e_m\}$, $m \in \mathbb{N}$ for the solution space. This can be done by separability of the solution space.
- Prove the existence and uniqueness of $u_m \in V_m$ such that

$$a(u_m, v_m) = \langle f, v_m \rangle \quad \forall v_m \in V_m, \quad (1)$$

by matrix formulation and positive definiteness of the formulated matrix.

Galerkin Approximation (Continued)

Sketch of the Procedure (Continued)

- To prove the uniform boundedness of the approximate solutions using Poincaré and Cauchy-Schwartz inequality.
- To show that the weakly convergent subsequence of u_m converges to the solution of the problem (1).
- To prove the uniqueness by Poincaré inequality.

Method

After showing existence and uniqueness of the weak solutions, the process consists of the following steps:

- **Triangulation ($\mathcal{T}_h$):** Partition of the domain Ω into triangular elements K of maximum diameter h .
- **Finite Element Space:** Construct $V_h \subset H_0^1(\Omega)$ such that $\dim(V_h) < \infty$, consisting of piecewise linear polynomials on each triangle.
- **Basis Functions (η_j):** Associated with each node j , construct linear polynomials η_j , where $\eta_j(x_k, y_k) = \delta_{jk}$.
- **Sparsity:** η_j only has support on elements sharing node j . The integral $\int_{\Omega} \nabla \eta_j \cdot \nabla \eta_k \, dx$ is zero for non-adjacent nodes, leading to a highly sparse stiffness matrix.

Error Calculation

After finding the approximate solution, we have to calculate the interpolation error between V_h and $H_0^1(\Omega)$.

Céa's Lemma

There exists a constant $C > 0$, which does not depend on h , such that

$$\|u - u_h\| \leq C \inf_{v_h \in V_h} \|u - v_h\|.$$

Error Analysis (Continued)

Sketch of Proof

- For ellipticity constant $\alpha > 0$ of the bilinear form $a(.,.)$ defined before,
$$\alpha \|u - u_h\|^2 \leq a(u - u_h, u - v_h + v_h - u_h) = a(u - u_h, u - v_h) + a(u - u_h, v_h - u_h).$$
- Prove that $a(u - u_h, v_h - u_h) = 0$.
- Using the continuity of the bilinear form $a(.,.)$, the inequality is obtained.

Simulation 1: Square Domain

Model Problem

Let $\Omega = (0, 2) \times (0, 2)$. Consider the following PDE,

$$\begin{cases} -\Delta u = 2(y(2-y) + x(2-x)) & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega. \end{cases}$$

Solution & Observation

- **Exact Solution:** $u(x, y) = xy(2-x)(2-y)$.
- **Observation :** The domain is convex, and the exact solution is a smooth polynomial. The FEM is expected to achieve optimal convergence rates.

Exact vs. Numerical Solution for Square Domain

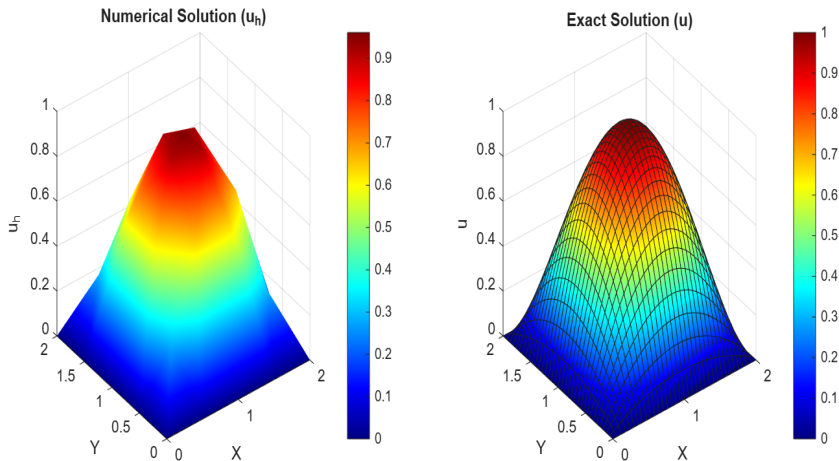


Figure: Comparison between exact solution (u) and fem induced solution (u_h) for $h = 0.5$.

Error Calculation for Square Domain

Mesh Size h	Max Absolute Error	L^2 -Error	H^1 -Error
0.5	4.779×10^{-2}	9.648×10^{-2}	5.968×10^{-1}
0.3	1.171×10^{-2}	3.750×10^{-2}	3.745×10^{-1}
0.1	1.924×10^{-3}	4.156×10^{-3}	1.246×10^{-1}
0.05	4.811×10^{-4}	1.004×10^{-3}	6.111×10^{-2}

- Errors systematically decrease as the mesh is refined ($h \rightarrow 0$).

Simulation 2: L-Shaped Domain

Model Problem

Let $\Omega = (0, 2) \times (0, 2) \setminus [1, 2) \times [1, 2)$. Consider the following PDE,

$$\begin{cases} -\Delta u = f & \text{in } \Omega, \\ u = 0 & \text{on } \partial\Omega, \end{cases}$$

where, $f : \Omega \rightarrow \mathbb{R}$ be a given function.

Exact Solution & Choosing $f : \Omega \rightarrow \mathbb{R}$

- **Challenge:** The re-entrant corner at $(1, 1)$ causes a singularity of the solution.
- **Exact Solution Chosen:**

$$u(x, y) = x(2 - x)y(2 - y)(1 - x)(1 - y) \cdot ((x - 1)^2 + (y - 1)^2)^{5/3}.$$

In the above problem, $f : \Omega \rightarrow \mathbb{R}$ is given as the Laplacian of the exact solution.

Exact vs. Numerical Solution for L-Shaped Domain

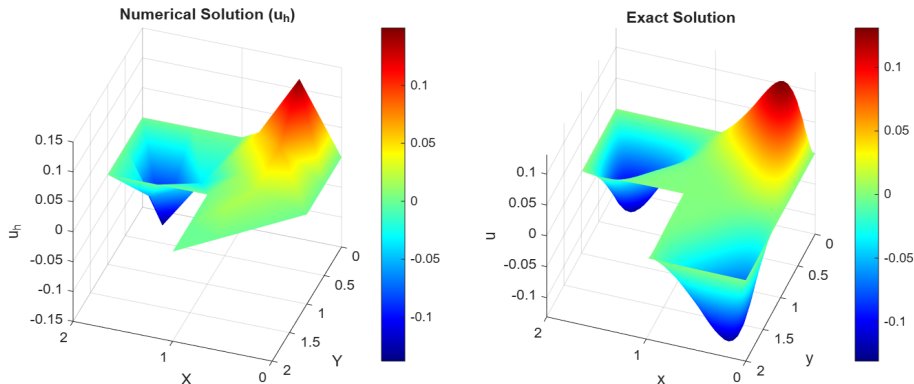


Figure: Comparison between exact solution (u) and numerical solution (u_h) for $h = 0.5$.

Error Calculation for L-Shaped Domain

Mesh Size h	Max Absolute Error	L^2 -Error	H^1 -Error
0.5	4.208×10^{-2}	3.896×10^{-2}	4.407×10^{-1}
0.3	4.338×10^{-2}	2.639×10^{-2}	2.937×10^{-1}
0.1	4.420×10^{-2}	1.607×10^{-2}	1.275×10^{-1}
0.05	4.427×10^{-2}	1.576×10^{-2}	8.438×10^{-2}

- The H^1 and L^2 errors decrease, but much slower than in the square domain.
- Noticeably, the Maximum Absolute Error stays around 4.4×10^{-2} despite mesh refinements, reflecting the influence of the re-entrant corner.

Comparison of Convergence Rates (EOC)

Experimental Order of Convergence (EOC) formula:

$$\text{EOC} = \ln(E_{h_i}/E_{h_{i+1}})/\ln(h_i/h_{i+1})$$

Mesh Refinement	Square Domain (H^1)	L-shaped Domain (H^1)
0.50 $\rightarrow$ 0.40	0.91	1.20
0.40 $\rightarrow$ 0.30	0.88	0.48
0.30 $\rightarrow$ 0.20	1.06	0.77
0.20 $\rightarrow$ 0.10	0.96	0.75
0.10 $\rightarrow$ 0.09	1.06	1.05
0.09 $\rightarrow$ 0.08	0.94	0.68

- The Square Domain maintains optimal convergence (≈ 1).
- The L-shaped domain fluctuates and displays suboptimal convergence rates due to re-entrant corner.

Conclusion

- The numerical experiments demonstrate both the effectiveness of the finite element method for square domains.
- Changing of domain geometry alters convergence behaviour of the numerical solutions.
- This reflects the requirement of the implementation of adaptive mesh refinements for non-convex domains.

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Thank You!